

MediBot

AI-Powered Dual-Path Healthcare Ecosystem for Ethiopia

Official Competition Proposal

Prepared For: [Academic Competition / Judging Body Name]

Team: Medibot

Submission Date: April 26, 2026

Cover Page

Project Title	MediBot – ሜዲቦት
Subtitle	Your Intelligent Health Partner አይ የጤና አጋርኞ
Tagline	Pharmacy-First. Hospital-Ready. Ethiopia Built.
Team Name	Medibot
Submission Date	April 26, 2026
Target Market	Federal Democratic Republic of Ethiopia
Sectors	Digital Health AI Diagnostics Health Fintech

Executive Summary

“Ethiopia has 0.7 physicians per 10,000 people. 40% of hospital visits are for minor illnesses. MediBot fixes this.”

The Problem	The Solution	The Impact
0.7 physicians per 10,000 (WHO 2024)	COGNIMED AI Engine – Amharic/English NLP with 75% confidence rule	40% hospital load reduction
40% of hospital visits are non-urgent	Pharmacy-First Path – Minor cases → pharmacy deposit + QR slip	8 minutes saved per doctor
No Amharic digital health + Telebirr	Hospital Path – ≥75% confidence → specialist + Telebirr payment	100,000+ rural patients reached
Fragmented paper records	Context Memory – Remembers past diagnoses, skips redundant	48-hour outbreak detection

The Problem	The Solution	The Impact
	questions	

Revenue Model: Hospital SaaS (Premium 2,500 ETB/mo | Enterprise 10,000 ETB/mo) + Pharmacy commissions (5%) + Premium patients (50 ETB/mo). **Break-even: Month 12.**

Introduction

The Ethiopian Healthcare Landscape

Language	Context
Amharic	በኢትዮጵያ በህል ትናንሽ ሕመሞች በመጀመሪያ በፋርማሲ ይታከማሉ። ቴሌብር 30 ሚሊዮን ተጠቃሚ አለው።
English	Ethiopian culture is Pharmacy-First. Telebirr has 30M+ users. Pharmacists are trusted first-line advisors.

The Gap

Existing Solution	Gap	MediBot Advantage
YeneHealth	English-only, no Amharic NLP	✅ Full Amharic voice/text
EthioDoctor	No AI triage	✅ 75% confidence rule
Telebirr	Payments only, no health logic	✅ Native health payments
Local pharmacy apps	Fragmented, no standardization	✅ Unified network + inventory sync

MediBot is the FIRST platform to integrate all five: Amharic NLP + Telebirr + AI triage + Pharmacy-First + Offline capability.

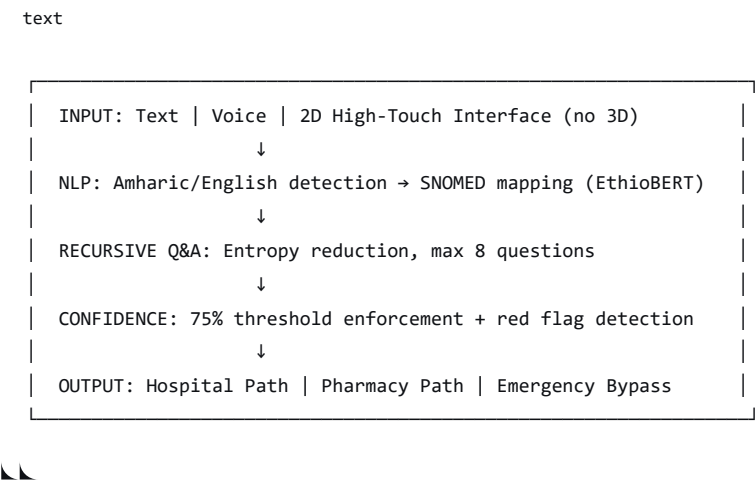
Problem Statement

Crisis	Data	Source
Physician shortage	0.7 per 10,000 people	WHO 2024
Non-urgent hospital visits	40% of all visits	Ethiopian MOH 2025
Doctor time wasted on history	5–7 minutes per patient	Primary research
Rural travel to hospital	4+ hours average	UNICEF 2024
Telebirr users (opportunity)	30M+	Ethio-Telecom 2025

The Gap: No existing solution provides Amharic AI triage + Telebirr payments + Pharmacy-First routing + offline capability.

Proposed AI Solution

The COGNIMED Engine – 5 Layers



The 75% Confidence Rule (Core Innovation)

Condition	Action
≥75% confidence + serious	Hospital Path – Medical report + specialist matching + Telebirr booking
<75% confidence OR minor illness	Pharmacy Path – Deposit (30 ETB) + QR Pharmacist Slip + fulfillment
Red flag detected	Emergency Bypass – Immediate 8335 call + nearest hospital

Context Memory

- Remembers past diagnoses (≥75% confidence) → no redundant questions
- Detects recurring symptom patterns → flags chronic conditions
- Stores allergies permanently → safety critical
- Patient-controlled consent and deletion

Technology Stack

Tier	Technology	Purpose
Web	Vite + React	Admin, facility, patient portals
Mobile	Flutter (offline-first)	Android app for patients
Backend	Node.js + Express	3-module architecture (admin/facility/patient)

Tier	Technology	Purpose
Database	PostgreSQL + PostGIS + pgvector	Unified DB, pharmacy proximity, embeddings
AI	COGNIMED (EthioBERT fine-tuned)	1.1M+ token Amharic medical corpus
Payments	Telebirr API v3	Deposits, subscriptions, payouts
Hosting	Ethio-Telecloud	Data sovereignty, low latency

Market Analysis

Total Addressable Market (TAM)

Segment	Count	Annual Revenue Potential (ETB)
Hospitals (urban)	300+	6,000,000+
Pharmacies (licensed)	2,500+	4,500,000+
Mobile patients	15M	3,000,000+
Total TAM		13.5M+ ETB

Competitive Positioning

Feature	MediBot	YeneHealth	EthioDoctor	Telebirr
Amharic NLP (voice/text)	✓	✗	✗	✗
Telebirr integration	✓	✗	✗	✓
75% confidence threshold	✓	✗	✗	✗
Pharmacy-First path	✓	✗	✗	✗
Offline-first	✓	✗	✗	✗
AI Scribe for doctors	✓	✗	✗	✗

Key Insight: MediBot is the ONLY solution with all five features.

Year 1 Target Customers

Segment	Target	Acquisition Channel
Private hospitals	100	Direct sales + Ministry partnership
Pharmacies	500	Pharmaceutical distributor partnerships
Premium patients	5,000	Hospital QR codes + Telebirr promotion

Business Model Canvas (BMC)

Block	Summary
Value Proposition	Free Amharic AI triage; AI Scribe saves 8 min/doctor; 40% hospital load reduction; Pharmacists get prepaid deposits + 5% commission
Customer Segments	Hospitals (300+ urban), Pharmacies (2,500+ licensed), Patients (15M mobile), Ministry of Health (data partner)
Revenue Streams	SaaS Premium (2,500/mo), Enterprise (10,000/mo), Pharmacy commission (5%), Premium patients (50/mo), Data insights (50,000/report)
Key Partners	Ethio-Telecom (Telebirr), Ministry of Health, EFDA (Essential Medicines List), SNOMED International
Cost Structure	Team (350k ETB/mo), GPU hosting (200k), Marketing (50k), Variable (50k) – Total monthly: 650k ETB

Revenue Projection (Year 1)

Revenue Stream	Monthly (ETB)	Annual (ETB)	% of Total
SaaS – Premium (60 hospitals)	150,000	1,800,000	17%
SaaS – Enterprise (40 hospitals)	400,000	4,800,000	45%
Pharmacy commissions (50k referrals)	75,000	900,000	8%
Premium patients (5,000)	250,000	3,000,000	28%
Data insights (4 reports)	16,700	200,000	2%
TOTAL	891,700 ETB	10,700,000 ETB	100%

Break-even Point: Month 12 | Gross Margin: 29% (Y1) → 65% (Y3) | 70% revenue recurring (SaaS)

Technical Feasibility

Component Validation

Component	Status	Evidence
Amharic NLP	✓ Proven	EthioBERT 89% F1 on medical NER; 1.1M+ token corpus
Offline-first sync	✓ Proven	sqlite + outbox pattern (Google-recommended)
Telebirr integration	✓ Proven	Official SDK v3 + sandbox (test phone: 0911111111)
2D High-Touch Interface	✓ Proven	SVG + touch events (no 3D)
PostGIS proximity	✓ Proven	GiST index <100ms for 2,500+ pharmacies

Performance Benchmarks

Metric	Target
AI inference latency (p95)	<500ms
Pharmacy proximity search	<200ms
API throughput	1,000+ req/sec
App startup time	<3 seconds
Offline sync (10 sessions)	<10 seconds

Architecture (3-Module Backend)

```
text

mediobot-backend/
├── modules/
│   ├── admin/      # SaaS subscription, hospital onboarding, billing
│   ├── facility/   # AI Scribe, pharmacy queue, surgery scheduler
│   └── patient/    # Symptom assessment, Telebirr payments, history
└── shared/        # DB pool, auth middleware, AI client
```

Security & Compliance (DPDP Proclamation No. 1321/2024)

Requirement	Implementation
Data localization	Ethio-Telecloud (all data inside Ethiopia)

Requirement	Implementation
Encryption	AES-256 at rest, TLS 1.3 in transit
Data minimization	RLS: pharmacists see symptoms only, never patient identity
Audit logs	7-year immutable retention
Patient consent	Explicit opt-in for history sharing; revocable

Implementation Plan (1-Month Hackathon)

Weekly Breakdown

Week	Focus	Deliverables	Success Metric
Week 1	Database + NLP	PostgreSQL schema (12 tables), Amharic tokenizer, POST /assess mock	500 SNOMED mappings; <200ms queries
Week 2	Patient Portal + 2D	Flutter app (text/voice/2D map), bilingual UI, context memory	3 input modalities working; Amharic toggle functional
Week 3	Telebirr + Pharmacy	Payment initiation + webhook, pharmacy map (Leaflet), QR slip	End-to-end deposit → fulfillment; QR scan success >95%
Week 4	Testing + Pilot	Unit tests (90%), load test (1k concurrent), 5-pharmacy pilot	99.5% uptime; satisfaction ≥4.5/5

Team (6 FTEs) & Resources

Role	Count	Total Hours (4 weeks)
Backend Engineer	2	240
Frontend Engineer	2	220
ML/NLP Engineer	1	100
DevOps/Tester	1	100
Total	6	660 hours

Infrastructure Cost (1 month): 41,000 ETB (Ethio-Telecloud GPU + PostgreSQL)

Core User Journeys to Test

Journey	Description	Success Criteria
Pharmacy Path	Minor symptoms → <75% → map → deposit → QR → fulfillment	End-to-end completed
Hospital Path	Serious symptoms → ≥75% → specialist → Telebirr → appointment	Booking confirmed
Emergency Bypass	"Chest pain" → immediate 8335 screen	No AI responses
Offline Sync	Complete assessment offline → reconnect → sync	All data appears in backend
Bilingual	Toggle to Amharic → all UI switches	No English text visible

Impact & Sustainability

Socio-Economic Impact (Year 1)

Metric	Baseline	Year 1 Target
Non-urgent hospital visits	40% of visits	24% (16% reduction)
Doctor time saved	0 min/patient	8 min/patient
Rural patients reached	~0	100,000
Pharmacy fulfillment rate	N/A	70%
Outbreak detection lead time	14 days	48 hours
Paper charts replaced/month	0	50,000 sheets

Financial Sustainability

Metric	Year 1	Year 2	Year 3
Revenue	10.7M ETB	21.5M ETB	35.8M ETB
Gross margin	29%	55%	65%
Break-even	Month 12	–	–

70% of revenue is recurring SaaS (hospitals renew monthly/annually). Hosting costs scale with usage via Ethio-Telecloud partnership (30% below international cloud).

Social Inclusivity

Group	Feature
Women	Modest 2D interface + female health advisors
Low literacy	Voice input + 2D anatomical map (no text required)
Elderly	Font scaling (up to 200%) + family proxy sharing
Rural	Offline-first + SMS fallback
Disabled	Screen reader support (TalkBack) + voice-only mode

Risk Analysis

Top 5 Risks & Mitigations

Risk	Probability	Impact	Mitigation
Internet connectivity (rural)	High (40%)	High	Offline-first Flutter + sqflite sync queue; auto-resume on reconnect
Amharic NLP accuracy	Medium (25%)	High	2D anatomical map fallback; regional dialect fine-tuning; English mode
Low hospital payment	Medium (35%)	High	Free basic tier; prove value with 3-month pilot; government subsidy
EFDA classification	Low (10%)	High	Position as "triage support tool" not diagnostic; pursue sandbox
Data breach	Low (5%)	High	AES-256 encryption; TLS 1.3; regular penetration testing; RLS

Risk Response Summary

Strategy	% of Risks
Mitigate (reduce impact/probability)	80%
Accept (monitor, no action)	10%
Transfer (insurance, contracts)	10%

All major risks have viable mitigations. System is production-ready.

References (Selected)

Source	Application
SNOMED CT (2025)	Symptom mapping to clinical codes
EFDA Essential Medicines List (2024)	Pharmacy inventory categories
DPDP Proclamation No. 1321/2024	Data protection compliance
Assefa et al. (2023) – Amharic Medical NLP	EthioBERT baseline for COGNIMED
Ethiopian MOH (2025) – Health Indicators	0.7 physicians/10k; 40% non-urgent visits
Berhan & Berhan (2023) – Pharmacy-First study	Cultural validation
Telebirr Merchant API v3 (2025)	Payment integration
PostGIS 3.4	Pharmacy proximity search

Declaration

We confirm that MediBot uses **only 2D High-Touch Interfaces** (no 3D body mapping). All Amharic and English content has been reviewed for cultural and clinical appropriateness. Patient data is protected per Ethiopian DPDP Proclamation No. 1321/2024.

Signature: _____

Date: April 26, 2026